

Problem-Solution Fit Canvas

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Canvas Element	Details
1. Customer Segment(s) [CS]	Residents, farmers, and local disaster-management officials in flood-prone regions.
2. Problems / Pains + Frequency [PR]	Lack of a fast, personalized way to check flood risk — frequent concern, especially during monsoon season.
3. Triggers to Act [TR]	Onset of heavy or unusual rainfall, monsoon season alerts, recent flood news in the region.
4. Emotions Before / After [EM]	Before: anxiety and uncertainty. After (with a quick prediction): reassurance, or informed urgency to prepare.
5. Available Solutions – Pros & Cons [AS]	Government flood bulletins (slow, generic); TV/radio warnings (not location-specific); manual meteorological forecasting (labor-intensive, delayed).
6. Customer Limitations (Budget, Devices) [CL]	Limited access to expensive real-time weather sensors; varying digital literacy; basic internet-enabled devices only.
7. Behavior + Intensity [BE]	High-intensity, manual searching of news/social media for flood updates during heavy rainfall periods.
8. Channels of Behavior [CH]	Online: web search, social media, government portals. Offline: local announcements, community word-of-mouth.
9. Problem Root / Cause [RC]	No accessible, instant, data-driven tool exists that lets an individual enter local weather conditions and get an immediate flood-risk assessment.
10. Your Solution [SL]	Rising Waters — a Machine Learning-based web application that takes rainfall and weather inputs and instantly predicts flood risk using a trained Random Forest model.